

HELIOSYNC SYSTEMS

# HelioSync I10 Hybrid

Three-phase solar inverter - technical manual

|                 |                             |
|-----------------|-----------------------------|
| Model           | I10 Hybrid                  |
| Product code    | HS-I10-3P                   |
| Rated AC output | 10 kW                       |
| Edition         | Technical manual 3.0 - 2026 |

Design, installation, grid-interface, monitoring and compatible energy-management information for the HelioSync I10 Hybrid.

Read the identification, safety and applicability information before relying on this document.

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## 3. PV DC input

### PV strings

Connect only photovoltaic strings within the stated MPPT voltage and current limits. String design is separate from EV charging capacity.

### Isolation

Provide DC isolation and surge protection as required by the installation design.

### No vehicle connection

PV DC conductors must not be connected to a vehicle or wallbox. Conversion and protection are required before energy reaches the AC distribution system.

|                              |              |
|------------------------------|--------------|
| MPPT voltage range           | 180-850 V DC |
| Maximum DC voltage           | 1,000 V DC   |
| MPPT trackers                | 2            |
| Maximum recommended PV array | 15 kWp       |

## 4. AC output

### Rated output

The inverter provides up to 10 kW three-phase AC output under rated conditions.

### AC bus

In normal grid-connected operation, inverter output joins the household AC distribution bus. Loads may receive a mixture of PV energy and grid energy.

### EV charging implication

A wallbox set to 11 kW can operate while PV output is below 11 kW when site rules permit grid import. The inverter rating is not the same as a wallbox output rating.

|                            |                                      |
|----------------------------|--------------------------------------|
| Rated AC power             | 10 kW                                |
| Nominal voltage            | 3-phase 400 V                        |
| Nominal frequency          | 50 Hz                                |
| Maximum continuous current | 15.2 A per phase                     |
| Power factor range         | Configurable within grid-code limits |

## 5. Grid connection

### Point of connection

Connect the inverter to the building distribution system through the designed protective devices and isolator.

### Anti-islanding

The inverter disconnects from the grid when required by its certified protection profile.

### Metering

A compatible three-phase smart meter measures import, export and household load for energy-management functions.

### Backup output

The optional backup output is not the normal supply point for the installed EV wallbox unless a separate design explicitly approves it.

## 6. Energy management

### Measurements

The controller uses inverter generation, grid import/export and household load measurements.

### Export limitation

A configured export limit reduces PV output when export would exceed the setpoint.

### Flexible loads

Compatible flexible loads can receive a power or current setpoint based on available surplus and the household connection limit.

### Data quality

If meter data is invalid, the controller applies its fail-safe behaviour and should not infer surplus from inverter output alone.

## 7. EV charging integration

### Compatible wallbox

The HelioSync gateway supports the GridNest W11 through Modbus TCP profile HS-EV-2.

### Power allocation

The energy manager calculates a wallbox current setpoint from PV generation, household load, grid limit and the configured import policy.

### Grid-assisted mode

With grid import enabled, the W11 may request up to its configured 11 kW while the grid supplies the shortfall after available PV contribution.

### Surplus-only mode

With grid import disabled, the controller reduces or pauses charging so net import remains near the configured threshold.

|                    |                                         |
|--------------------|-----------------------------------------|
| Wallbox profile    | HS-EV-2 / GridNest W11                  |
| Transport          | Modbus TCP                              |
| Update interval    | 5 seconds nominal                       |
| Grid import option | Enabled or disabled by installer policy |

## 8. Battery interface

### Compatible battery

Use only listed high-voltage storage systems and approved firmware.

### Priority

The operating mode determines whether PV energy serves household load, battery charging, export or flexible loads first.

### Not the EV battery

The inverter battery interface is for stationary storage. The vehicle battery is managed through the AC wallbox and the vehicle onboard charger.



## 9. Operating modes

### Self-consumption

PV serves household load first, then charges stationary storage, then exports subject to limits.

### Time-of-use

The controller can schedule stationary battery charge and discharge within configured windows.

### EV priority

A flexible-load priority can allocate available power to the wallbox while respecting the grid and household limits.

### Manual mode

Installer testing may use a fixed setpoint. Restore the intended automatic policy after commissioning.

## 10. Alarms and records

### Grid alarm

Check voltage, frequency and site protective devices before reset.

### Meter communication

A missing meter prevents reliable import/export control and flexible-load allocation.

### Insulation alarm

Isolate PV strings and obtain qualified inspection.

### Event log

The local event log records time, code, operating mode and selected measurements. It does not by itself prove an insurance event.

| Code | Meaning                           |
|------|-----------------------------------|
| G102 | Grid voltage outside limit        |
| M210 | Smart-meter communication lost    |
| D330 | PV insulation resistance low      |
| E410 | Energy-manager target unavailable |

# 11. Installation and commissioning

## Qualified design

String voltage, fault current, protective devices, cable sizes and grid settings must be designed and tested by qualified personnel.

## Device identity

Record the inverter model and serial number in the site connection schedule.

## Functional tests

Confirm meter direction, phase mapping, export control, wallbox communication and grid-import policy.

## Handover

Give the owner the inverter manual, wallbox manual, site schedule and commissioning record as one technical set.

## 12. Technical summary

### Applicability

Use the site connection schedule to confirm that this is the installed inverter.

### Source control

Source ID s6-inverter-datasheet; technical manual 3.0.

| Item                         | Specification           |
|------------------------------|-------------------------|
| Model                        | HelioSync I10 Hybrid    |
| Product code                 | HS-I10-3P               |
| Rated AC output              | 10 kW, three-phase      |
| PV input                     | 2 MPPT, 180-850 V range |
| Grid interface               | 3-phase 400 V, 50 Hz    |
| EV integration               | HS-EV-2 over Modbus TCP |
| Compatible installed wallbox | GridNest W11            |
| Direct vehicle connection    | Not supported           |